RF Fingerprint Dataset

Objective 3 | Acquisition inventory, signal validation, metadata and demonstration outputs

Scope assessment: This is a validated, reproducible two-transmitter subset built from the user-provided updated RF_Dataset.zip. It is not a completed four-transmitter, controlled-SNR dataset.

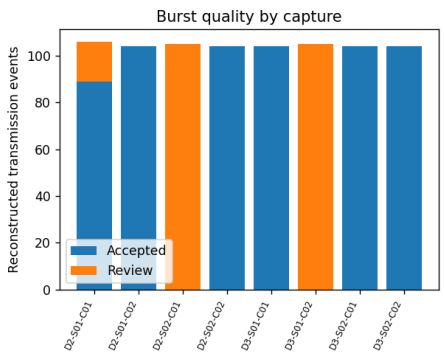
Summary of measured work

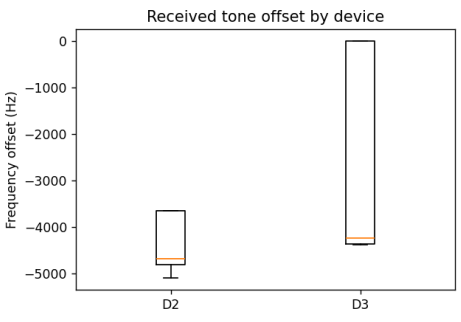
Metric	Result	Meaning
Original MATLAB captures	8	Two devices (D2 and D3), two sessions per device
Source detector bursts	879	Originally saved burst-array column counts
Reconstructed candidate events	836	Re-detected from original raw I/Q
Primary accepted observations	609	Passes conservative event and capture QC
Quarantined for review	227	Retained with reason, raw I/Q, and provenance
Optional class-balanced view	594	297 genuine events per class; no duplication

Capture-by-capture quality audit

Capture	Stored	Rebuilt	Accepted	Review
D2/S01/C01	106	106	89	17
D2/S01/C02	104	104	104	0
D2/S02/C01	105	105	0	105
D2/S02/C02	104	104	104	0
D3/S01/C01	104	104	104	0
D3/S01/C02	148	105	0	105
D3/S02/C01	104	104	104	0
D3/S02/C02	104	104	104	0
Total	879	836	609	227

Source recordings are unchanged, retained in original_recordings.zip. Rebuilt events are not a claim that all original detector bursts were valid.





Processing method and feature definitions

Signal and acquisition basis

ADALM-PLUTO over-the-air at a declared 2.4 GHz center frequency, 1 MS/s baseband, a 100 kHz complex single-tone signal, nominal 1,024-sample burst plus 4,096-sample gap, fixed distance 50 cm, and 35 dB manual receiver gain. The receiver E(4) assignment is user-declared and not independently verified from old captures. Only D2 and D3 have genuine source recordings in this release.

Re-extraction rules | RAW_REEXTRACT_V1_CORE48_960

The pipeline reads complex raw_iq, smooths its magnitude over 32 samples, and thresholds midway between the 10th and 90th amplitude percentiles. Gaps of up to 300 samples between detected active spans are merged. A nominal accepted event must have an active span of 900–1,150 samples, at least one neighboring onset separated by 5,120 ± 20 samples, a full 1,024-sample extraction window, finite features, and FFT peak within 85–115 kHz.

A capture with verified overflow events but unknown affected frame positions is conservatively quarantined as a whole. No original recording is discarded, re-labeled as a different transmitter, or synthetically enlarged.

Five consistently recomputed fingerprint features

Feature	Operational definition	Units
F1 — Transient energy	Mean $\|IQ\|^2$ over the first 200 aligned samples	ADC-relative magnitude²
F2 — Mean amplitude	Mean $\|IQ\|$ over stable-core samples 49–960	ADC-relative magnitude
F3 — Amplitude variance	Sample variance of stable-core $\|IQ\|$	ADC-relative magnitude²
F4 — Residual phase variance	Sample variance of unwrapped phase after linear detrending of core	rad²
F5 — Tone offset	Linear-phase-slope frequency minus 100,000 Hz	Hz

These features are newly extracted using one consistent definition, rather than concatenating incompatible previously saved feature matrices. F5 includes transmitter–receiver reference mismatch and must not be described as isolated transmitter-only CFO.

Quality issues found

D3/S01/C02: 148 source detector bursts are inconsistent with 104 regularly spaced transmission periods; raw pulse activity is about 1,920 samples per period, not the nominal 1,024. The pipeline reconstructs 105 candidates including an initial boundary fragment and quarantines them all.

D2/S02/C01: Reported six receiver overflow events, without sample-level attribution; all 105 candidates quarantined. D2/S01/C01: 17 of 106 rebuilt events fail duration or cadence QC.

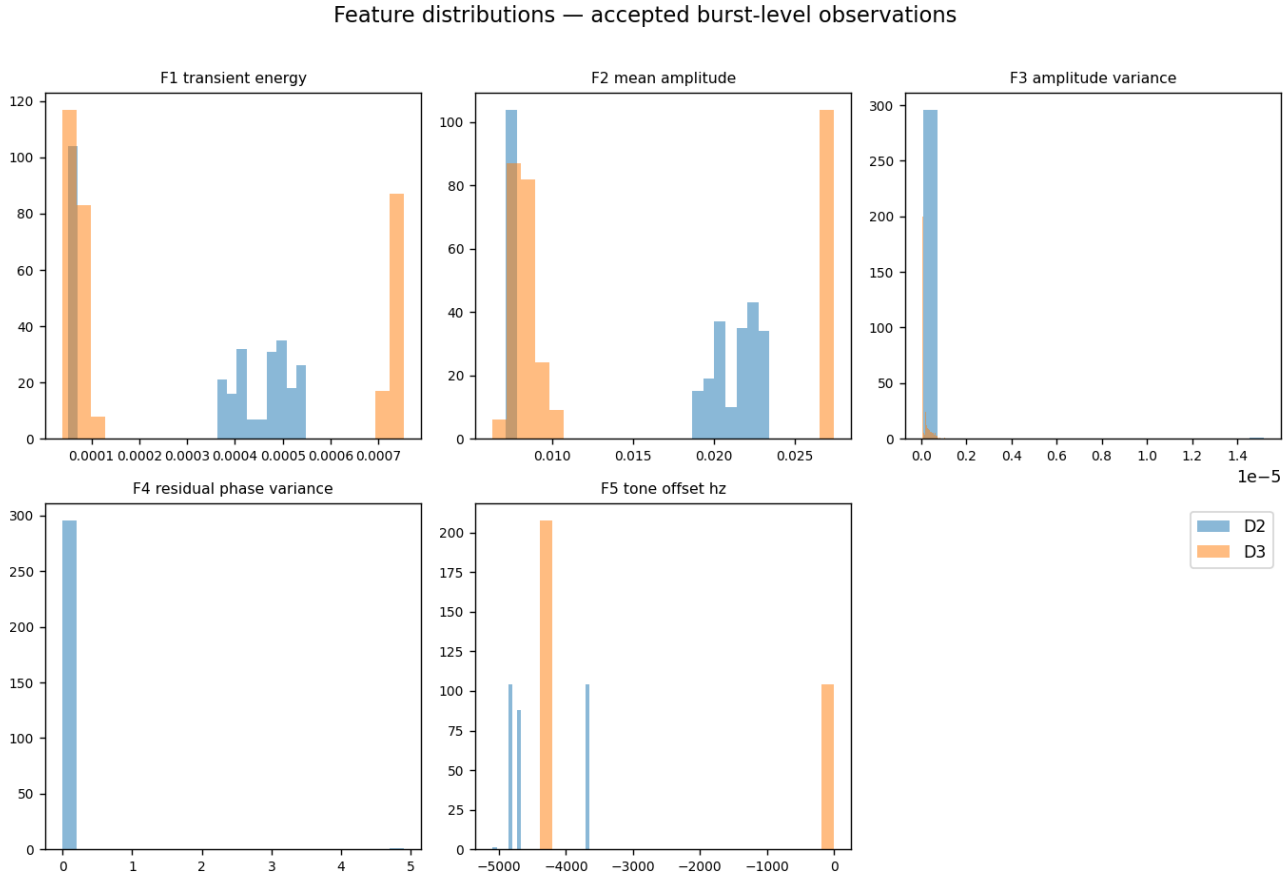
Metadata and SNR policy

Actual recorded dates supplied via MATLAB are included with provenance. D2/S01/C02 has an inferred session date; D3/S02 capture dates could not be verified and remain blank. Missing TX gains and overflow counts are not replaced with guesses. The calculated burst-to-gap power contrast is not calibrated measured SNR; the measured_snr_db column is intentionally blank and controlled SNR conditions are unverified.

Feature assessment and holdout evaluation

The following statistical work and classifier results are supplementary demonstration diagnostics. They do not establish that Objective 4 authentication works. Bursts from the same recording are highly correlated and must not be randomly mixed across training and testing.

Exploratory feature behavior



Feature distributions may reflect transmitter identity, but also vary by session, capture gain, power level, channel conditions and oscillator state. A feature priority ranking is provided in the CSV as exploratory, training-session-only variance ratios; it is not validated device-specific importance.

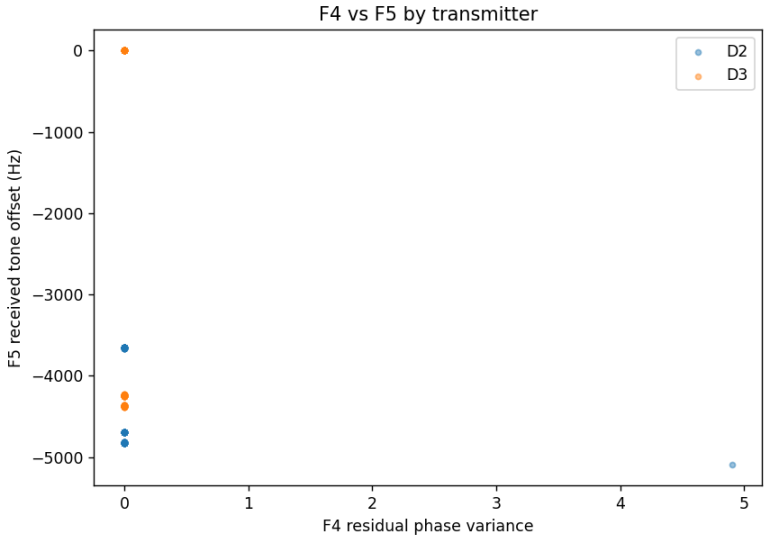
Grouped holdout: S01 training vs S02 testing

Training data: 297 accepted events. Test data: 312 accepted events. A single distinct session per class is held out; standardization is fit exclusively on the training portion. No random burst-level leakage.

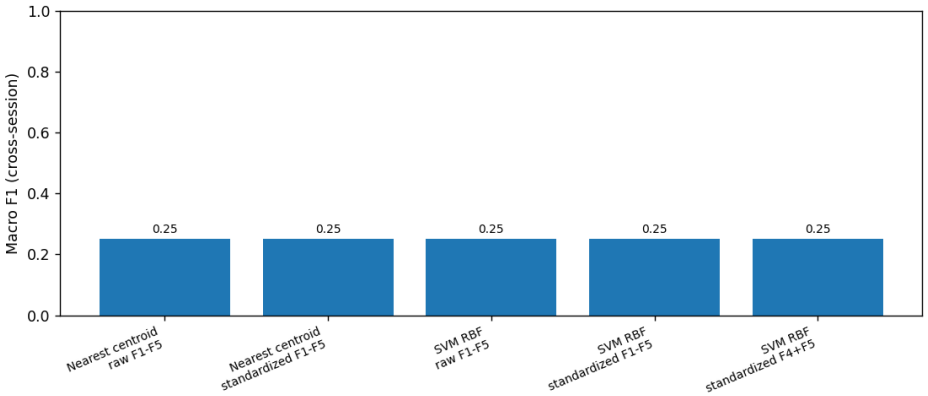
Model	Test accuracy	Macro F1
Nearest centroid — raw	33.33%	0.250
Nearest centroid — standardized	33.33%	0.250
SVM RBF — raw	33.33%	0.250
SVM RBF — standardized	33.33%	0.250
SVM RBF — F4+F5 standardized	33.33%	0.250

For every tested model, the test confusion matrix (true rows D2, D3; predicted columns D2, D3) is [[104, 0], [208, 0]]. All D3 test events are misclassified as D2. This is worse than a credible cross-session identification system and indicates severe generalization failure under this split.

Visual diagnostics and decisions



F4 vs F5 for accepted observations. Apparent global class structure must not be interpreted independently of the strong session effects.



Phase accounting and next work

Project stage	Outcome
Phase 0 — experiment definition	Two known transmitter classes; fixed receiver declared; full four-TX objective open
Phase 1 — master dataset/validation	609 accepted and 227 review events; metadata and audit preserved
Phases 2–4 — feature statistics/scaling	Exploratory plots, feature priority, standardized tests recorded
Phases 5–7 — baseline and SVM	Executed; cross-session performance unsatisfactory, not success
Phase 8 — visualization	Distribution, F4/F5 and PCA figures produced
Phases 9–10 — authentication/final assessment	Not implemented or validated as complete authentication

To finish Objective 3, gather at least four distinct transmitters with a fixed receiver, collect actual calibrated SNR regimes across multiple dates, verify all capture metadata, repair/re-record quarantined sessions, then re-run grouped validation. A separate fixed receiver may require additional hardware beyond the four Pluto units previously available.

Reproduce all outputs: python rf_dataset_builder.py original_recordings.zip reproduced_output. Every event is linked to its original file and source SHA-256 in the master/audit CSVs. See README.md and DATA_DICTIONARY.md for exact field definitions.